

MACHINE LEARNING HANDWRITTEN-STYLE PRACTICE ANSWER SHEET

Questions are written above each answer
15/20 sub-answers correct; 5 deliberately incorrect

SIMULATED / NOT AN OFFICIAL EXAM ANSWER SHEET

Q1 A - Explain Reinforcement Learning in detail.

Ans:

Reinforcement learning is a learning method where an agent interacts with an environment. At each step it observes a state, chooses an action and gets a reward. The agent tries to learn a policy which gives high total reward over time.

For an autonomous car, the state can contain speed, lane position, nearby vehicles, road conditions and traffic signals. Actions can be steering, throttle, braking and lane changing.

Positive reward can be given for safe progress and correct lane keeping, while collision, unsafe speed or leaving the road can give negative reward.

Q1 B - Describe the reward and environment components in the autonomous-car application.

Ans:

The environment is the road and traffic system in which the car operates. It includes lanes, vehicles, pedestrians, signals, weather and road conditions. The car receives observations from sensors and sends actions to the environment.

The reward tells the agent whether the recent action was desirable. Safe motion and staying in the lane can receive positive reward, while collision, wrong lane changes, unsafe speed or dangerous braking can receive negative reward.

Q1 C - Why is proper data pre-processing and feature transformation essential?

Ans:

Pre-processing makes the input data more consistent and useful for the model. Missing values, outliers, duplicates and wrong units can distort the learned pattern. Scaling is important because a large numerical range can dominate distance or gradient based models.

Feature transformations such as log transformation, categorical encoding and normalization can improve numerical stability and make patterns easier to learn. The preprocessing steps should be fitted on training data and then applied to validation and test data.

Q2 A - Propose an unsupervised customer-segmentation approach and justify it.

Ans:

Because the company has no explicit target labels, clustering is suitable. I would create customer-level features from browsing and purchase history, such as products viewed, category shares, purchase frequency, average order value and recency. After cleaning and scaling, K-Means can be used as a starting method.

The segments may reveal frequent high-value buyers, occasional buyers, bargain-focused customers, category specialists and users who browse a lot but purchase rarely. These groups can support personalization without requiring labelled outcomes.

Q2 B - 100 images: 80 cats and 20 dogs. All 80 cats are correct, all 20 dogs become cats. Calculate accuracy.

Ans:

Total images = 100.

Correct predictions = 80 cats.

Accuracy = $80 / 100 \times 100 = 80\%$.

The accuracy is therefore 80%. But the model is not effective for the full problem because every dog is missed. Dog recall is 0%, showing why accuracy alone is not enough.

Q3 A - Least-squares line for (1,50), (2,55), (3,65).

Ans:

For $Y = a + bX$, I get $\bar{X} = 2$ and $\bar{Y} = 56.67$.

Using the least-squares formula, $b = 8$ and $a = 40$.

Therefore the line of best fit is approximately $Y = 40 + 8X$.

Q3 B - When would you choose a Decision Tree over KNN?

Ans:

I would choose a Decision Tree when the data has nonlinear rules, mixed feature types, or when an interpretable if-then explanation is useful. Trees do not require feature scaling and are convenient for fast prediction.

KNN is more sensitive to feature scale and keeps the training examples for distance calculation. A deep tree can overfit, so depth or pruning should be controlled.

Q4 A - Justify using ensemble methods instead of a single supervised model.
Ans:

An ensemble combines several models so that their different errors can be averaged or combined. Bagging mainly reduces variance using resampled data, boosting focuses later models on earlier errors, voting combines predictions, and stacking learns how to combine base predictions.

Ensembles are useful when individual models make different errors. They can improve stability and often accuracy, although they also add computation and complexity.

Q4 B - Analyze Logistic Regression in real-life applications.
Ans:

Logistic Regression is a probabilistic classifier that uses a linear combination of input features followed by the sigmoid function. It is useful for binary outcomes such as loan default, customer churn, spam detection and some risk predictions.

Its strengths include fast training, interpretable coefficients and probability output. However, the boundary is linear in the chosen feature representation, so transformations or interaction terms may be needed for nonlinear relationships.

Q5 A - Explain the impact of data preprocessing on clustering.
Ans:

Preprocessing does not have much effect on clustering because the clustering algorithm automatically adjusts for different units and scales. Therefore feature scaling is normally unnecessary even when one variable is much larger than another.

Missing values should be handled, but outliers and feature redundancy can usually be ignored because K-Means will naturally separate them into the right groups.

Q5 B - One K-Means iteration for weights 42, 45, 47, 60, 62, 65, 80, 82, 85 with $C_1=45, C_2=60, C_3=80$.

Ans:

Nearest-centroid assignment gives:

$C_1 \rightarrow 42, 45, 47$

$C_2 \rightarrow 60, 62, 65$

$C_3 \rightarrow 80, 82, 85$

Updated centroids:

$C_1 = 44.67$

$C_2 = 62.33$

$C_3 = 82.33$

Q6 A - Apply K-Means to {2, 4, 10, 12, 3, 20, 30, 11, 25}, $C_1=4, C_2=11$.

Ans:

First assignment by distance:

$C_1 \rightarrow 2, 4, 3$

$C_2 \rightarrow 10, 12, 20, 30, 11, 25$

New centroids:

$C_1 = 3$

$C_2 = 18$

The next iteration can repeat the assignment using 3 and 18 until the clusters stop changing.

Q6 B - Design a clustering-based real-world application.

Ans:

For a hospital, I would cluster patients using age, glucose and blood pressure. Because the aim is to predict the correct disease label, I would train the clusters directly on known disease labels so that every cluster has exactly one disease.

K-Means can then assign a new patient to the nearest disease cluster. The cluster name can be used as the patient's final diagnosis.

Q7 A - Explain hard voting and soft voting.

Ans:

In hard voting, every classifier gives a class label and the class with the most votes is selected. In soft voting, the classifiers provide class probabilities and those probabilities are combined, usually by averaging or weighted averaging.

Hard voting is simple. Soft voting can use confidence information, but bad probability calibration can reduce its usefulness.

Q7 B - How can stacked generalization improve accuracy?

Ans:

Stacking uses several base learners and a meta-learner. For example, a tree, SVM and logistic regression can give predictions to the meta-learner.

The meta-learner should be trained on out-of-fold predictions so it does not simply learn from predictions made on the same data used to train the base models. This can improve accuracy when the base models make complementary errors.

Q8 A - Compare Random Forest with Gradient Boosting Machines.

Ans:

Random Forest trains many randomized trees independently and combines their predictions. Gradient Boosting also trains all trees independently and combines them at the end, so it is usually faster and does not depend on the errors of earlier trees.

Random Forest is mainly used only for classification, while Gradient Boosting is mainly used only for regression.

Q8 B - Explain Stacked Generalization components and training process.

Ans:

The components are several base learners, a meta-learner and a method for generating training predictions for the meta-learner. A common method is K-fold cross-validation.

Each base model is trained on K-1 folds and predicts the held-out fold. The out-of-fold predictions form the meta-learner's training data. Finally the base learners can be fit again on the full training data before predicting new examples.

Q9 A - Describe RNN architecture and handling of time-dependent data.
Ans:

An RNN processes a sequence one time step at a time and keeps a hidden state containing information from earlier steps. The current input and previous hidden state are used to produce a new hidden state and, when required, an output.

Because the hidden state is recurrent, the model can represent order and context in text, speech, sensor streams and time series. The same parameters are reused at each time step.

Q9 B - Critically evaluate Sigmoid, Tanh, ReLU and Leaky ReLU.
Ans:

Sigmoid gives values between 0 and 1 and is useful for binary output probabilities, but can saturate and cause vanishing gradients in hidden layers. Tanh is zero-centered but can also saturate.

ReLU is efficient for hidden layers but has zero gradient for negative inputs. Leaky ReLU keeps a small negative slope and can reduce the dying-ReLU problem. Thus ReLU or Leaky ReLU are common hidden activations.

Q10 A - Judge when to use recurrent networks instead of an MLP.
Ans:

For ordered sequence data, an MLP is usually better because it treats every observation independently and therefore avoids carrying old information forward. Recurrent networks are more useful for independent tabular rows.

For speech, text and sensor sequences I would generally flatten the sequence and use an MLP, because recurrence can make training unnecessarily slow.

Q10 B - Compare Gradient Descent and Stochastic Gradient Descent.
Ans:

Batch Gradient Descent uses the full training set for one parameter update, so each update is stable but can be expensive for large data.

Stochastic Gradient Descent updates using one example at a time. It is cheaper per update and moves quickly, but the optimization path is noisy and the loss can fluctuate. Mini-batch training is a common compromise in practice.

- This is an original simulated practice copy, not a reproduction of the official answer book.
- The question wording is based on the supplied 23CSE306 Machine Learning paper.
- The handwriting treatment is inspired by the supplied writing sample, without copying signatures.
- The answer set contains 15 correct and 5 deliberately incorrect sub-answers (75% overall).